COVER LETTER

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Dear Hiring Manager,

I am writing to express my interest in the Software Developer position within the Robotics Department. As a graduate student in Computer Information Technology at Purdue University, I've developed a strong foundation in designing reliable and scalable software systems that serve real-world needs.

For me, software development goes beyond writing code — it's about understanding the complete flow of a system, how components interact, and how design decisions impact performance and user experience. I take pride in structuring solutions that are both efficient and maintainable, using clean architecture principles and a deep understanding of data flow, concurrency, and system reliability.

Throughout my academic and professional journey, I have learned to approach problems methodically — analyzing requirements, mapping dependencies, and translating complex logic into simple, modular systems. I'm especially drawn to environments that value collaboration, continuous improvement, and the impact of thoughtful engineering on user outcomes.

What excites me most about this opportunity is the chance to contribute to the Robotics Department's mission through technology. I see software not just as a tool, but as a bridge between human creativity and machine capability — something that perfectly aligns with your department's vision.

I would be thrilled to bring my problem-solving mindset and system design skills to your team. Thank you for considering my application. I look forward to the opportunity to discuss how I can contribute to your department's work.

Sincerely

Raghavendra Melam

RAGHAVENDRA MELAM

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Summary

Software Developer experienced in building secure, scalable backend systems with Java, Spring Boot, and REST APIs. Proven ability to collaborate with diverse teams to integrate responsive front-end features and develop production-grade applications. Demonstrates a strong focus on clean architecture, efficient coding, and practical problem-solving.

Purdue University **2023 - 2025**

M.S., Computer Information Technology

- **GPA:** 3.5/4.0

VVIT **2019 - 2023**

B.Tech, computer science & engineering

- **GPA:** 3.0/4.0
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Skills

- **Languages & Concepts**: Java (Core, 8–17)
- **Technologies**: JDBC, Jakarta EE
- **Frameworks**: Spring, Spring Boot, Spring Data JPA, Spring Web (MVC, REST), Spring Batch, Spring Mail, Spring Security 6.0
- **Databases**: Oracle, MySQL, MongoDB
- **Containerization & Cloud**: Docker, AWS, Kubernetes
- **Testing & Quality**: JUnit, Mockito, JaCoCo, SonarQube, JMeter
- **Web & Frontend**: HTML, CSS, JavaScript, Angular
- **DevOps & Tools**: Git, GitHub, Jenkins CI/CD, Maven, Postman, Swagger, IntelliJ, Eclipse
- **Agile & SDLC**: Agile, Scrum, Jira
- **Soft Skills**: Self-Starter, Strong Problem-Solving Skills, Collaboration, Time Management, Quick Learner, Adaptability to New Technologies, Excellent Verbal and Written Communication Skills

Projects

Health Insurance System (HIS)

- Built a modular microservices-based backend system to automate health insurance workflows for a simulated state-level platform. The app supports onboarding, eligibility checks, plan management, benefit issuance, and reporting of citizens.
- Designed 8 backend modules including User Management, Eligibility Determination, Benefit Issuance, and Reporting
- Built rule-based qualification logic for 5 government plans (Medicaid, Medicare, SNAP, CCAP, NJW)
- Developed secure RESTful APIs and applied Spring Security 6.0 with role-based access
- Used Java multithreading to optimize the Correspondence module (30% faster email processing)

- Applied TDD practices with JUnit, automated quality checks via SonarQube, and performance testing using JMeter
- Designed scalable MySQL schema and currently integrating Angular-based frontend

Crop Sales & Storage Management System

- Built a backend application for a South Indian Mirchi crop export company to digitize crop transactions, commission calculation, and cold storage billing.
- Created REST APIs for crop sales with real-time price entry, 3.5% commission logic, and automated billing
- Developed secure authentication and role-based access (Admin, Customers) using JWT
- Built a cold storage (AC build) module to handle timestamped crop entry, fee tracking, and invoicing
- Integrated JavaMailSender to notify customers with pricing and digital receipts
- Enabled dynamic search for customer records (name, date, crop sold) and built real-time calculation interface
- Conducted unit/integration tests with JUnit, applied SonarQube & JMeter, and deployed on AWS EC2 for 24/7 use

Real-Time Truck Tracking System

- Developed a **real-time GPS-based truck tracking system** to monitor shipment movements, enabling logistics efficiency similar to platforms used by **Amazon, FedEx, and UPS**.
- Built a **Kafka-based asynchronous architecture** to decouple services — GPS Simulator produced live location updates, consumed by Shipment Service for processing.
- Implemented **Kafka consumers and producers** to handle high-throughput location data ingestion and ensure scalable communicati on between microservices.
- Designed and exposed **RESTful APIs** for tracking shipment status, retrieving historical location data, and assigning drivers dynamically.
- Integrated **Notification Service** to alert clients via email upon successful shipment delivery, leveraging Kafka for event-driven triggers.
- Utilized **Spring Data JPA** for persistence and optimized queries to fetch real-time and historical tracking data efficiently.